Started on	Sunday, 24 July 2022, 3:36 PM
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Time taken	11 secs
Marks	0.00/111.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Which phenomenon is commonly used in a functional MRI scan to make inferences about brain activity?

Select one:

- Echo planar imaging (EPI)
- Diffusion imaging
- High angle low-frequency pulses
- Blood oxygen level dependent (BOLD) phenomenon
- Fast low angle shot radiofrequency pulses (FLASH)

Explanation:

The goal of fMRI is to visualise brain activity and so spatial resolution is sacrificed to the benefit of temporal resolution. fMRI is not at sufficient temporal resolution to directly measure the neuronal signal and so it measures an indirect index of neuronal activity – the blood oxygen-level dependent (BOLD) contrast. The principle behind BOLD is that an active population of neurons will receive more fresh, oxygen-risk blood flow than non-active neurons. It is this variation between the magnetic signal of oxygenated and deoxygenated blood in the brain that constitutes the basis for the fMRI signal.

Ref: Martinelli C, Shergill SS. Everything you wanted to know about neuroimaging and psychiatry, but were afraid to ask. BJPsych Advances. 2015 Jul;21(4):251-60.

The correct answer is: Blood oxygen level dependent (BOLD) phenomenon

Question **2**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a good practice while using interpreters?

Select one:

- Ask the interpreter about patient's guarded nature while answering
- Offering to debrief the interpreter if needed after interpretation
- Encourage interpretation and then translation of mental status examination
- Brief the interpreter before seeing the patient
- Sticking to the same interpreter for a single case

When using an interpreter, explain the need for providing a literal translation of a patient's account. If the interpreter provides a 'summarised translation' or 'interpretations' of the reported account, several salient aspects may be missed during a Mental Status Examination.

The correct answer is: Encourage interpretation and then translation of mental status examination

Question **3**

Not answered

Marked out of 1.00

Flag question

Which of the following is an advantage of open ended questions over closed questions?

Select one:

- Answers to open questions are highly time-saving
- Answers to open questions are doctor-directed
- Answers to open questions are highly reliable
- Answers to open questions are less precise
- Answers to open questions are more informative

Explanation:

An open question does not suggest the possible answer, while a close question expects a limited range of replies. Beginning an interview with open questions can provide informative answers, before turning to more closed questions to clarify details or factual points.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 50.

The correct answer is: Answers to open questions are more informative

Question **4**

Not answered

Marked out of 1.00

Flag question

In which of the following situations is closed questioning a useful technique?

Select one:

- In a highly distracted patient
- To interview an overtalkative patient
- At the later part of interview
- To rule out less likely symptoms
- All of the listed options

Closed questions are better avoided in early parts of interview as they can produce prescribed answers lacking in detail. Close questions must be avoided in highly suggestible patients. A good technique is to start with open; move to closed by the end of the interview. Useful to rule out less likely symptoms. The answers for closed questions can only be 'yes' or 'no', in most occasions. When clubbed with non-facilitative gestures, can be detrimental to interview process. Stating a presumption followed by tags can be very directive.

The correct answer is: All of the listed options

Question **5**

Not answered

Marked out of 1.00

Flag question

All of the following techniques are useful in facilitating disclosure EXCEPT

Select one:

- Silence
- Positive reinforcement
- Acknowledgement of affect
- Suggestive questions
- Validation

Suggestive questions have answers that are contained in the question itself. This misleads both the patient and the doctor. The patient is often left with little choice than to agree with the suggestion.

The correct answer is: Suggestive questions

Question **6**

Not answered

Marked out of 1.00

Flag question

The most common opportunistic infection in a patient with AIDS is

Select one:

- Leprosy
- Toxoplasma
- Cytomegalo virus
- Cryptococcus
- Syphilis

Toxoplasmosis is common in AIDS. Brain imaging shows multiple ring-shaped contrast enhancing lesions. MRI is considered superior to CT scanning with IV contrast material with either modality improving the diagnostic yield and accuracy.

The correct answer is: Toxoplasma

Question **7**

Not answered

Marked out of 1.00

Flag question

An anorexic patient repeatedly vomits. Which of the following is a possible complication?

Select one:

- Metabolic acidosis
- Hyponatraemia
- Hyperkalaemia
- Metabolic alkalosis
- None of the listed options

Explanation:

In anorexia nervosa, hypokalemic/hypochloraemic metabolic alkalosis can result from vomiting (loss of gastric acid). Metabolic acidosis can result from laxative abuse.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 414.

The correct answer is: Metabolic alkalosis

Question **8**

Not answered

Marked out of 1.00

Flag question

Which of the following SPECT finding is seen in Alzheimer disease?

Select one:

- Multiple zones of ischaemia in left hemisphere
- Increase in temporal perfusion
- Increase in parietal perfusion
- Increase in frontal perfusion
- Decrease in temporal perfusion

Explanation:

In Alzheimer disease, SPECT shows temporal and posterior parietal hypoperfusion.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 157.

The correct answer is: Decrease in temporal perfusion

Question **9**

Not answered

Marked out of 1.00

Flag question

Wernicke encephalopathy is directly caused by a deficiency of

Select one:

- Thiamine
- Alcohol dehydrogenase
- Riboflavin
- Niacin
- Aldehyde

Explanation:

Wernicke encephalopathy is an acutecondition characterised by **classic triad** of **confusion, ataxia, and ophthalmoplegia/nystagmus.** However, the triadonly seen in **10%** of cases ; confusion in **80%** of cases. It occurs secondary to **thiamine (vitamin B1) deficiency.**

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 606.

The correct answer is: Thiamine

Question **10**

Not answered

A patient undergoing an assessment for dementia is asked to draw a circle and place numbers inside the circle to depict the face of a clock. All of the following domains of cognition are crucial for

Marked out of 1.00

Flag question

successful completion of this task except

Select one:

- Executive function
- Constructional praxis
- Comprehension
- Orientation to time
- Visuospatial ability

Explanation:

The clock-drawing test does not assess one's orientation to time.

Ref: https://www.sralab.org/sites/default/files/2017-07/Clock%20Drawing%20Test%20Instructions.pdf

The correct answer is: Orientation to time

Question **11**

Not answered

Marked out of 1.00

Flag question

A previously reliable and conscientious man has become increasingly profane, irascible, and irresponsible over the past 3 - 6 months. These symptoms suggest that a tumour, if present, is most likely to be located in the

Select one:

- Medial wall
- Temporal lobe
- Parietal lobe
- Cerebellum
- Frontal lobe

Explanation:

Lesions of the frontal lobes are characterised by changes in personality. The frontal lobe syndrome, which is most commonly produced by trauma, infarcts, tumours, multiple sclerosis, or frontotemporal dementia, consists of slow thinking, poor judgement, decreased curiosity, social withdrawal, and irritability. Patients may display apathetic indifference to experience that suddenly explode into impulsive disinhibition.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 935.

The correct answer is: Frontal lobe

Question **12**

Not answered

Marked out of 1.00

Flag question

Alexia without agraphia (pure alexia) is seen in lesions of

Select one:

- Middle meningeal artery
- Posterior inferior cerebellar artery

- Middle cerebral territory
- Posterior cerebral territory
- Anterior cerebral territory

In alexia without agraphia or pure word blindness, patients retain the ability to write, but are unable to read (even words that they have just written) and often have right homonymous hemianopia. This disorder results from the loss of visual input to the language area [usually the dominant (left) angular gyrus] without the involvement of the language area or output from the language area to the motor cortex. The ability to process visual input into language (i.e. reading) is lost in these patients. The pathways involved in this syndrome are believed to be localized to the inferior portion of the splenium of the corpus callosum on the side of the dominant (usually left) cerebral hemisphere. This is associated with posterior cerebral artery damage.

The correct answer is: Posterior cerebral territory

Question **13**

Not answered

Marked out of 1.00

Flag question

During a clinical interview, a doctor tells her patient 'I am impressed by what you have achieved'. What type of statement is this?

Select one:

- Normalisation
- Counter-transference
- Validation
- Negative therapeutic alliance
- Statement of respect

Affirmative statements that are genuine and appropriate indicating respect and dignity offer positive reinforcement during a psychiatric interview.

The correct answer is: Statement of respect

Question **14**

Not answered

Marked out of 1.00

Flag question

All of the following help localizing a neurological lesion to cerebellum EXCEPT

Select one:

- Ankle clonus
- Hypotonia
- Dysdiadochokinesia
- Ataxia
- Past pointing

Cerebellar signs: Ataxia, hypotonia, dyssynergia (incoordination), dysmetria (past pointing), dysdiadochokinesis, dysrhythmia and dysarthria

The correct answer is: Ankle clonus

Question **15**

Not answered

Marked out of 1.00

Flag question

Various components of language that must be tested in order to localize the brain lesion causing aphasia include all EXCEPT

Select one:

- copying
- comprehension
- naming
- Repetition
- fluency

Reading, writing, repetition, fluency, comprehension and naming are tested to localize aphasia..

The correct answer is: copying

Question **16**

Not answered

Marked out of 1.00

Flag question

In occasional cannabis user, how long is cannabis detected in urine after last use?

Select one:

- 3 hours
- 4 weeks
- 12 hours
- 8 days
- 3 days

Explanation:

In the setting of occasional use, cannabis can be detected in the urine for up to 3 days.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 631.

The correct answer is: 3 days

Question **17**

Not answered

Marked out of 1.00

Flag question

A 61-year-old right-handed man presented to the emergency department with an acute onset of right-leg weakness. Soon his weakness progressed to include both arms and legs, and he stopped speaking. On examination, he was alert but produced no spontaneous speech or spontaneous movements. Which artery is most likely to be implicated in this presentation?

Select one:

- Cerebello pontine artery
- Posterior cerebral artery
- Anterior cerebral artery
- Middle cerebral artery

- Basilar artery

Explanation:

Damage to the anterior cingulate cortex is known to be a primary cause for akinesia and mutism and has frequently been described in case reports of patients with bleeding, stroke, or other types of severe brain injury. The most dramatic examples of akinesia and mutism follow bilateral lesions of the anterior cingulate cortex, such as after bilateral damage to the vascular supply of the anterior cerebral artery territory.

Ref: Arnts H, van Erp WS, Lavrijsen JC, van Gaal S, Groenewegen HJ, van den Munckhof P. On the pathophysiology and treatment of akinetic mutism. Neuroscience & Biobehavioral Reviews. 2020 May 1;112:270-8.

The correct answer is: Anterior cerebral artery

Question **18**

Not answered

Marked out of 1.00

Flag question

Inability to attend to the parts and the whole of an object at the same time is called

Select one:

- Somatognosia
- Visuospatial apraxia
- Prosopognosia
- Simultanagnosia
- Anosognosia

Explanation:

Simultanagnosia refers to impairment in the perception or integration of visual stimuli appearing simultaneously.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1144.

The correct answer is: Simultanagnosia

Question **19**

Not answered

Marked out of 1.00

Flag question

Select the most appropriate investigation for a patient on antipsychotic medications presenting with weight gain, excessive thirst and fatigue.

Select one:

- Fasting glucose
- Liver function tests
- Full blood count
- Electrocardiogram
- Fasting lipids

The symptoms of weight gain, excessive thirst and fatigue are suggestive of impaired glucose tolerance. A fasting blood glucose is indicated to diagnose antipsychotic related diabetes mellitus.

The correct answer is: Fasting glucose

Question **20**

Not answered

Marked out of 1.00

Flag question

A 24-year-old man who is being investigated for jaundice presents with a gradual change in personality characterised by disinhibition and recklessness. On examination, he has tremors and dysarthria. Which of the following signs is highly likely in this case?

Select one:

- Xanthomas on extensor aspect of forearm
- Golden brown pigmentation around cornea
- Port wine stain on face
- Ash leaf macules
- Cafe au lait spots

Explanation:

Wilson disease is a genetic disorder of copper metabolismthat leads to copper accumulation in various organs, primarily the liver and brain. Clinical symptoms are a result of organ dysfunction due to the direct or indirect effects of **copper accumulation**. It is typically described as a **hepato-neurological disease**, with focus on signs of liver disease and extrapyramidal symptoms (e.g. tremor, dystonia, dysarthria). **Psychiatric symptoms** include personality disturbances (e.g. antisocial behaviour, irritability, disinhibition), mood abnormalities (e.g. bipolar, depression, suicide attempts), psychosis, and cognitive impairment. **Kayser-Fleischer rings** (green-brown copper deposits on Descemet's membrane of the cornea) are present in 95% of patients with neurological symptoms, in 50-60% of patients without neurological symptoms, and in only 10% of asymptomatic siblings.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 166.

Litwin T et al. Psychiatric manifestations in Wilson's disease: possibilities and difficulties for treatment. Therapeutic advances in psychopharmacology. 2018 Jul;8(7):199-211.

The correct answer is: Golden brown pigmentation around cornea

Question **21**

Not answered

Marked out of 1.00

Flag question

The most appropriate description for executive function is

Select one:

- The capacity to identify stimulus that induces a perception
- The capacity of problem solving
- The capacity of controlling emotion and impulse
- The capacity of purposefully applying one's physical skill
- The capacity of accomplishing goal-directed behaviour

Explanation:

As conceptualised by Muriel Lezak, the executive functions include volition (i.e. formulation of a goal, motivation to achieve the goal, and awareness of one's own ability to achieve the goal), planning,

purposive action (response selection and initiation, maintenance, switching, and stopping), and execution (which involves self-monitoring and self-correction as well as control of the spatiotemporal aspects of the response).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 50.

The correct answer is: The capacity of accomplishing goal-directed behaviour

Question **22**

Not answered

Marked out of 1.00

Flag question

A 29-year-old man presents with multiple abdominal scars from different surgeries, giving an appearance of checker board abdomen. What is the most likely psychiatric cause?

Select one:

- Conversion disorder
- Delusional disorder
- Hypochondriasis
- Dissociative disorder
- Factitious disorder

Explanation:

In factitious disorder, patients intentionally falsify their symptoms and past history and fabricate signs of physical or mental disorder, with the primary aim of obtaining medical attention and treatment. This patient's repeated presentation to a surgical ward (probably with various pain symptoms leading to surgeries) is suggestive of a factitious disorder.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 876.

The correct answer is: Factitious disorder

Question **23**

Not answered

Marked out of 1.00

Flag question

A forensic inpatient is asked to undergo narcoanalysis by the Crown Court. Which of the following statements about narcoanalysis is NOT true?

Select one:

- Useful technique when repression and dissociation are suspected.
- Organic muteness gets better with amobarbital infusion
- Psychogenic muteness gets better with amobarbital infusion
- Amobarbital is used
- Benzodiazepines can be used

Interviews with amobarbital have both diagnostic and therapeutic indications. These are sometimes helpful in differentiating nonorganic and organic conditions, particularly in patients with symptoms of catatonia, stupor, and muteness. Organic conditions tend to worsen with infusions of amobarbital, but nonorganic or psychogenic conditions tend to get better because of disinhibition, decreased anxiety, or increased relaxation. Therapeutically, amobarbital interviews are useful in disorders of repression and dissociation such as amnesia and fugue. Benzodiazepines can be substituted for amobarbital.

The correct answer is: Organic muteness gets better with amobarbital infusion

Question **24**

Not answered

Marked out of 1.00

Flag question

Which of the following is characteristic of a senile pupil?

Select one:

- Sluggish light and accommodation reaction
- Sluggish accommodation reaction; preserved light reaction
- Exaggerated accommodation and light reaction
- Preserved accommodation reaction; sluggish light reaction
- No changes in light and accommodation reaction

The pupil size becomes smaller with aging. This constricting of pupil is called senile miosis; the pupil becomes sluggish to light and accommodation.

The correct answer is: Sluggish light and accommodation reaction

Question **25**

Not answered

Marked out of 1.00

Flag question

In a motorbike accident, a 21-year-old driver sustains significant head injury. Following recovery from emergency neurological status, it is found that he has significant cognitive deficits. Which of the following is NOT a test of parietal lobe functioning?

Select one:

- Two point discrimination
- Right left orientation
- Lexical fluency
- Astereognosia
- Finger agnosia

Explanation:

Tests of parietal lobe functioning include finger agnosia, astereognosia, dysgraphaesthesia, two-point discrimination, and right-left orientation. Although of disputed clinical value, Gerstmann syndrome is classically described as right-left disorientation, finger agnosia, dysgraphia, and dyscalculia, due to a lesion of the dominant (usually left) parietal lobe.

The correct answer is: Lexical fluency

Question **26**

Not answered

Marked out of 1.00

Flag question

Which of the following cognitive domains is assessed by Wisconsin Card Sorting Test?

Select one:

- Language
- Visuospatial skills
- Immediate recall
- Retrieval

- Planning and set shifting

Explanation:

The Wisconsin card sorting task is a test of frontal lobe functioning. The patient has to determine the rule for card allocation and allocate cards accordingly. When the rule changes, a patient with frontal lobe dysfunction is likely to make more errors (tests response inhibition and set shifting).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 60.

The correct answer is: Planning and set shifting

Question **27**

Not answered

Marked out of 1.00

Flag question

When copying a complex figure, a patient with brain injury manages to reproduce global features of the design but misses the finer details of the figure. He is most likely to have

Select one:

- Angular gyrus lesion
- Bilateral damage
- Right hemisphere damage
- Fusiform gyrus lesion
- Left hemisphere damage

Explanation:

Distinctive qualitative errors in constructing block designs and in drawing a complex geometric configuration (e.g., Rey-Osterrieth Complex Figure test) can be seen with either right or left hemisphere damage. In the presence of lateralised damage to the right hemisphere, impaired performance often reflects the patient's inability to appreciate the "Gestalt" or global features of a design. In contrast, damage to the left hemisphere commonly results in inaccurate reproduction of internal details of the design, but the Gestalt is more likely to be preserved.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 48-49.

The correct answer is: Left hemisphere damage

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following is a risk factor for QT prolongation on ECG?

Select one:

- Hypokalemia
- Age 25 to 45
- Male sex
- Hypercalcemia
- Tachycardia

Explanation:

Risk factors for QTc prolongation include hypokalaemia, hypomagnesaemia, hypocalcaemia, arrythmias, bradycardia, extremes of age, and female gender.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. pp. 26-27.

The correct answer is: Hypokalemia

Question **29**

Not answered

Marked out of 1.00

Flag question

In MMSE, intersecting pentagons test primarily assesses which of the following functions?

Select one:

- Constructional praxis
- Language
- Fluency
- Visual retention
- Orientation

Explanation:

Praxis refers to a complex cognitive function, in which the motor system is used to execute complex learned actions and movements, such as the copying of drawings. Constructional praxis declines with age and can become impaired by several neurodegenerative diseases. Usual tasks to evaluate this function include the copying of drawings or drawing three-dimensional shapes. In the MMSE, praxis is evaluated by drawing two intersecting pentagons, in which the interconnected area should be shaped like a rhombus. Praxis functions depend mainly on the parietal lobe, which is specialised for visuospatial information, visual attention, body position, and whole body object perception relative to self.

Ref: Martinelli JE et al. Performance of the Pentagon Drawing test for the screening of older adults with Alzheimer's dementia. Dementia & neuropsychologia. 2018 Mar;12(1):54-60.

The correct answer is: Constructional praxis

Question **30**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a good technique to use for a poor historian?

Select one:

- Using verbal and nonverbal facilitation
- Collecting information in parts over multiple interviews
- Shifting to the neutral ground when necessary
- Using open-ended questions
- Using multiple choice questions

Use of open-ended questions and commands to increase the flow of information, use of continuation techniques to keep the flow coming and shifting to the neutral ground when necessary are useful techniques for a poor historian.

The correct answer is: Using multiple choice questions

Question **31**

Not answered

Marked out of 1.00

Flag question

A lactate solution is infused in a 32-year-old woman. This results in acute anxiety, palpitations, sweating, fear of loss of control and dizziness. What is the most likely diagnosis?

Select one:

- Lactic acidosis
- Malingering
- Generalized anxiety disorder
- Panic disorder
- Lactose intolerance

Explanation:

The biological basis of panic disorder has been postulated to be aberrant metabolic activity induced by lactate, from studies involving exercise-induced panic attacks and replicated by IV lactate infusion.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 370.

The correct answer is: Panic disorder

Question **32**

Not answered

Marked out of 1.00

Flag question

The presence of intracytoplasmic inclusion bodies is a microscopic finding in which of the following conditions?

Select one:

- Alcoholic dementia
- Lewy body dementia
- Alzheimer disease
- Creutzfeldt-Jakob disease
- Huntington disease

Explanation:

Pathologically, dementia with Lewy bodies is characterised by the presence of Lewy bodies – intraneuronal inclusions consisting mainly of alpha-synuclein aggregates and ubiquitin – in the neocortex, forebrain, brainstem, and other parts of the nervous system.

Ref: Mueller C et al. The prognosis of dementia with Lewy bodies. The Lancet Neurology. 2017 May 1;16(5):390-8.

The correct answer is: Lewy body dementia

Question **33**

Not answered

Marked out of 1.00

Which of the following is a compound question?

Select one:

- I think you are not well. Am I right?

Flag question

- What can I do for you?
- What is the average height of British women?
- Do you know how IQ is calculated?
- Do you go on holidays and are you able to relax then?

Adding two or more questions in a single statement is referred to as a compound question. This confuses the patient and may lead to either a vague response or no response.

The correct answer is: Do you go on holidays and are you able to relax then?

Question **34**

Not answered

Marked out of 1.00

Flag question

Which of the following tasks is most useful in evaluating a patient's ability to concentrate?

Select one:

- Naming five large cities
- Reporting current world events
- 3 items recall
- Performing serial 7s
- Estimating the distance between Sheffield and London

Serial Sevens is a clinical test that forms a part of the Mini Mental Status Examination; it is a quick and easy test of concentration and memory, used especially if there is a clinical suspicion of cognitive dysfunction.

The correct answer is: Performing serial 7s

Question **35**

Not answered

Marked out of 1.00

Flag question

One of the most well established anatomical changes seen in schizophrenia is

Select one:

- Enlargement of the lateral ventricle
- Enlargement of corpus callosum
- Enlargement of the amygdala
- Enlargement of the third ventricle
- Enlargement of cerebellar peduncles

Explanation:

Brain structural abnormalities are widely reported in schizophrenia with large-scale meta-analyses detecting a smaller hippocampal volume in patients compared to controls, followed by a smaller amygdala, thalamus, nucleus accumbens, and intracranial volumes, as well as larger pallidum and larger lateral ventricle volumes.

Ref: Keshavan MS, Collin G, Guimond S, Kelly S, Prasad KM, Lizano P. Neuroimaging in schizophrenia. Neuroimaging Clinics. 2020 Feb 1;30(1):73-83.

The correct answer is: Enlargement of the lateral ventricle

Question **36**

Not answered

Marked out of 1.00

Flag question

The symptom or sign that responds earlier than others when thiamine is replaced in a patient with Wernicke's encephalopathy is

Select one:

- Opthalmoplegia
- Apathy
- Confusion
- Ataxia
- Amnesia

Ophthalmoplegia responds within hours, but cognitive impairment takes longer.

The correct answer is: Opthalmoplegia

Question **37**

Not answered

Marked out of 1.00

Flag question

Which of the following arteries on occlusion leads to symptoms suggestive of dorsolateral prefrontal dysfunction?

Select one:

- Middle cerebral
- Peduncular
- Basilar
- Posterior cerebral
- Anterior cerebral

DLPFC consists of the lateral portions of Brodmann areas 9 - 12, areas 45, 46, and the superior part of area 47. These regions receive their blood supply from the middle cerebral artery.

The correct answer is: Middle cerebral

Question **38**

Not answered

Marked out of 1.00

Flag question

Copying intersecting polygons in MMSE tests the functions of

Select one:

- Hippocampus
- Parietal lobes
- Temporal lobes
- Cerebellum
- Frontal lobes

Explanation:

Praxis refers to a complex cognitive function, in which the motor system is used to execute complex learned actions and movements, such as the copying of drawings. Constructional praxis declines with age and can become impaired by several neurodegenerative diseases. Usual tasks to evaluate this function include the copying of drawings or drawing three-dimensional shapes. In the MMSE, praxis is evaluated by drawing two intersecting pentagons, in which the interconnected area should be shaped like a rhombus. Praxis functions depend mainly on the parietal lobe, which is specialised for visuospatial information, visual attention, body position, and whole body object perception relative to self.

Ref: Martinelli JE et al. Performance of the Pentagon Drawing test for the screening of older adults with Alzheimer's dementia. Dementia & neuropsychologia. 2018 Mar;12(1):54-60.

The correct answer is: Parietal lobes

Question **39**

Not answered

Marked out of 1.00

Flag question

Which of the following statements is true concerning bedside tests of orientation?

Select one:

- Most patients do not remember the length of time spent in hospital in spite of intact orientation
- Orientation to time is often affected in delirium
- Disorientation to one's own name indicates hippocampal lesions
- Intact orientation excludes memory disorders
- Orientation to person is the most sensitive test

Explanation:

Orientation is usually assessed to time, place and person; it is not particularly sensitive, and intact orientation does not exclude a significant memory disorder, particularly if there is concern about memory from an informant. Time orientation is the most helpful, and should include the time of day. Many normal people do not know the exact date, and being out by two days or less is considered normal when scoring this formally. Time intervals are often poorly monitored by patients with delirium, moderate to severe dementia, and in the amnesic syndrome, and are easily tested by asking about the length of time spent in hospital. Person orientation includes name, age, and date of birth. Disorientation to one's own name is usually only seen in psychogenic amnesia.

Ref: Cognitive assessment for clinicians -- Kipps and Hodges 76 .., http://jnnp.bmj.com/content/76/suppl_1/i22.full (accessed May 6, 2021)

The correct answer is: Orientation to time is often affected in delirium

Question **40**

Not answered

Marked out of 1.00

Flag question

The reason for using confrontation technique during a clinical interview is to

Select one:

- Deal with difficult patients
- Reveal some absurd notions that patients might hold
- Make patients stop lying to the clinician
- Help the patient face a difficult aspect of their problems
- Clarify certain associations that patients may not see

Confrontation technique points out to a patient something to which the doctor thinks the patient is missing or denying. This must be done in a respectful way with an aim to help patients face a difficult aspect rather than dismissing patients by pointing out a negative aspect.

The correct answer is: Help the patient face a difficult aspect of their problems

Question **41**

Not answered

Marked out of 1.00

Flag question

Amphetamines could be detected in urine for up to

Select one:

- 48 hours
- 30 days
- 6 hours
- 24 hours
- 7 days

Explanation:

Amphetamines can be detected in the urine for up to 48 hours.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 631.

The correct answer is: 48 hours

Question **42**

Not answered

Marked out of 1.00

Flag question

A girl attending your outpatient unit is 165 cm tall and weighs 50kg. What is her body mass index?

Select one:

- 20 to 21
- 16 to 17
- 17 to 18
- 18 to 19
- 19 to 20

Explanation:

Body mass index (BMI) is a ratio between weight and height and is more useful for predicting health risks than weight alone. BMI = Weight (in kg)/height (in m)^2.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 410.

The correct answer is: 18 to 19

Question **43**

Not answered

Marked out of 1.00

Flag question

Unilateral frontal lobe damage can result in

Select one:

- no spastic paresis
- contralateral spastic hemiplegia
- contralateral hypotonia
- ipsilateral spastic hemiplegia
- paraplegia

Damage to upper motor neurons originating in frontal motor strip can result in contralateral spastic hemiparesis.

The correct answer is: contralateral spastic hemiplegia

Question **44**

Not answered

Marked out of 1.00

Flag question

Which of the following conditions, when suspected, will necessitate performing a neurological procedure called Hoover's test?

Select one:

- Depression
- Conversion disorder
- Somatisation disorder
- Munchausen syndrome
- Hypochondriasis

Hoover's test is useful to identify patients with conversion disorder, who has paralysis with One or more limbs or one side of the face or body may be affected. Here the patient is unable to raise the affected limb from the couch but can raise the unaffected limb against resistance with demonstrable pressing down of the heel on the affected side. Ref: Oxford handbook of psychiatry; 746

The correct answer is: Conversion disorder

Question **45**

Not answered

Marked out of 1.00

Flag question

Patients with schizophrenia are least likely to perform poorly on which of the following tests?

Select one:

- Category test
- Trail B of the Halstead-Reitan battery
- Wisconsin card sort test
- Speech sounds perception test
- MMSE

Patients with schizophrenia perform far more poorly than controls on category test (assess higher-order

reasoning), Wisconsin card sort test (tests higher-order reasoning), speech sounds perception test, Trail B of the Halstead-Reitan battery (tests the ability to shift flexibility between cognitive sets). Tests of overall cognitive ability such as MMSE are usually intact in schizophrenia.

The correct answer is: MMSE

Question **46**

Not answered

Marked out of 1.00

Flag question

Which of the following signs can be elicited by having the patient extend both arms with the wrists dorsiflexed and palms facing forward and eyes closed?

Select one:

- Chorea
- Myoclonus
- Asterixis
- Athetosis
- Pronator drift

Asterixis can be elicited by having the patient extend both arms with the wrists dorsiflexed and palms facing forward and eyes closed. Brief jerky downward movements of the wrist are considered a positive sign. Asterixis is commonly seen with metabolic encephalopathies.

Ref: Neurological History and Physical Examination - Medscape, http://emedicine.medscape.com/article/1147993-overview (accessed May 5, 2021)

The correct answer is: Asterixis

Question **47**

Not answered

Marked out of 1.00

Flag question

The wavelet of EEG that is greater than 13 Hz in frequency is

Select one:

- Delta
- Beta
- Theta
- Gamma
- Alpha

Explanation:

Alpha waves: frequency range 8-13 Hz; highly rhythmic and constitute the dominant brain wave frequency of the normal eyes-closed awake EEG

Beta waves: frequencies that are faster than the upper 13 Hz limit of the alpha rhythm; not uncommon in average adult waking EEG studies, particularly over the frontal-central regions

Delta waves: frequency of 3.5 Hz or less; not present in the normal waking EEG, but a prominent feature of deeper stages of sleep; presence of significant generalised or focal delta waves in the awake EEG is strongly indicative of a pathophysiological process

Theta waves: frequency 4-7.5 Hz; small amount of sporadic, arrhythmic, and isolated theta activity can be seen in many normal waking EEG studies, particularly in frontal-temporal regions; a prominent

feature of the drowsy and sleep tracing; excessive theta in awake EEG, generalised, or focal, suggests the operation of a pathologic process

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 998-999.

The correct answer is: Beta

Question **48**

Not answered

Marked out of 1.00

Flag question

Failure of object recognition despite normal sensory input and motor functions is called

Select one:

- Visual agnosia
- Prosopagnosia
- Astereognosis
- Anosognosia
- Neglect

Explanation:

Anosognosia: inability to recognise a physical deficit in oneself (e.g. patient denies paralysed limb)

Visual agnosia: inability to recognise objects

Prosopagnosia: inability to recognise familiar faces that is not due to impaired visual acuity or level of consciousness

Astereognosis: inability to recognise familiar objects by touch

Neglect: failure to detect visual or tactile stimuli or to move the limb in the contralateral hemispace

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 49, 1136, 1143, 1145.

The correct answer is: Visual agnosia

Question **49**

Not answered

Marked out of 1.00

Flag question

In schizophrenia, the anatomical structure in the brain that is most consistently reduced in size is

Select one:

- striatum
- insula
- hypothalamus
- cerebellum
- hippocampus

Most of the listed structures have been observed to have structural abnormalities in schizophrenia. In fact, the hippocampus has been repeatedly shown to have a reduced volume though not in all studies. Insula, which forms a part of the paralimbic cotex, has been unequivocally shown to have volumetric abnormalities in schizophrenia. Other consistently affected structures include anterior cingulate cortex, dorsolateral prefrontal cortex and thalamus.

The correct answer is: insula

Question **50**

Not answered

Marked out of 1.00

Flag question

Which of the following laboratory tests should be conducted prior to initiating lithium treatment?

Select one:

- Alkaline phosphatase
- Thyroid-stimulating hormone (TSH)
- B12
- Erythrocyte sedimentation rate
- Calcium

Explanation:

Prior to commencing lithium therapy, conduct the following:

- Physical examination
- Full blood count
- Urea and electrolytes
- Thyroid function tests
- Renal function
- Baseline weight and height (body mass index)
- If clinically indicated: ECG, pregnancy test

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 350.

The correct answer is: Thyroid-stimulating hormone (TSH)

Question **51**

Not answered

Marked out of 1.00

Flag question

In MRS, which nuclei are commonly used for in vivo measure of phosphocreatinine?

Select one:

- H-1
- Na-23
- Li-7
- F-19
- P-31

Explanation:

Whereas routine MRI detects hydrogen nuclei to determine brain structure, magnetic resonance spectroscopy (MRS) can detect several odd-numbered nuclei. The readout of an MRS device is usually in the form of a spectrum, such as those for phosphorous-31 and hydrogen-1 nuclei. The MRS of the hydrogen-1 nuclei is best at measuring N-acetylaspartate (NAA), creatine, and choline-containing molecules, but can also detect glutamate, glutamine, lactate, and myo-inositol. MRS of phosphorous-31 can be used to determine the concentrations of phosphorous-containing compounds (e.g. ATP,

GTP, phosphocreatinine).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 71-72.

The correct answer is: P-31

Question **52**

Not answered

Marked out of 1.00

Flag question

A patient has normal accommodation reflex but no reaction on testing light reflex. What is the likely diagnosis?

Select one:

- Marcus Gunn pupil
- Argyll-Robertson pupil
- Holmes Adie pupil
- Hutchison pupil
- Horner pupil

Explanation:

An Argyll-Robertson pupil is a pupil of the eye that manifests accommodation by constricting when watching a finger that is brought close to the face but does not respond to light by constricting, a neurological sign of damage to the midbrain relays of the oculomotor nerve. In 1869, Douglas Argyll-Robertson described several patients whose pupils reacted poorly to light with a normal near response. Later the aetiology of this pupillary anomaly was discovered to be tertiary syphilis. Currently, diabetes leads the list of causes for the so called Argyll-Robertson Pupil or ARP.

Ref: Colman AM. Oxford Dictionary of Psychology, 4 th Edition. 2015.

The correct answer is: Argyll-Robertson pupil

Question **53**

Not answered

Marked out of 1.00

Flag question

A 32-year-old school teacher is admitted for constipation and acute abdominal pain. She develops motor weakness of her legs on the administration of hypnotics and diclofenac. She experiences visual and tactile hallucinations with intense anxiety. Which of the following laboratory test is indicated?

Select one:

- Serum folate
- Urine porphyrins
- Serum lipid levels
- Serum ceruloplasmin
- Urine glucose

The scenario depicts acute intermittent porphyria (AIP). It is one of the groups of disorders of haem metabolism, characterised by neurological and psychiatric manifestations without obvious cutaneous markers. AIP manifests itself by abdomen pain, neuropathies, and constipation, but, unlike most types of porphyria, patients with AIP do not have a rash. It is an autosomal dominant disorder with the presentation starting between ages 18 and 40. It is episodic in nature and the episodes are often triggered by certain medications including estrogens, barbiturates and benzodiazepines. Diclofenac can

precipitate an episode. Psychiatric manifestations include depression, anxiety, delirium and psychosis. Most important lab test is demonstrating increased urinary porphobilinogen during acute attacks. Treatment is aimed at reducing haem synthesis by administering haemin

The correct answer is: Urine porphyrins

Question **54**

Not answered

Marked out of 1.00

Flag question

Which one of the following has been used in vitro as a powerful method of analytical chemistry and enables the measurement of the concentration of different metabolites in the brain?

Select one:

- Structural MRI scan
- CT scan
- Functional MRI scan
- X-ray
- Magnetic resonance spectroscopy

Explanation:

Magnetic resonance spectroscopy enables information to be obtained about the biochemistry of tissues, which can be compared to identify tissue change following strokes or tumours. It enables detection of differences that may precede structural changes detectable with MRI.

Ref: Martinelli C, Shergill SS. Everything you wanted to know about neuroimaging and psychiatry, but were afraid to ask. BJPsych Advances. 2015 Jul;21(4):251-60.

The correct answer is: Magnetic resonance spectroscopy

Question **55**

Not answered

Marked out of 1.00

Flag question

Mrs Y is an elderly lady with cognitive impairment and visual hallucinations. Select the best imaging technique to diagnose her condition.

Select one:

- CT scan
- PET scan
- DAT scan
- f-MRI scan
- MRI Scan

Explanation:

Reduced dopamine transporter (DAT) uptake in basal ganglia demonstrated by SPECT or PET imaging is an indicative biomarker of dementia with Lewy bodies. The utility of DAT imaging in distinguishing dementia with Lewy bodies from Alzheimer dementia is well-established, with sensitivity (78%) and specificity (90%).

Ref: McKeith IG et al. Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. Neurology. 2017 Jul 4;89(1):88-100.

The correct answer is: DAT scan

Question **56**

Not answered

Marked out of 1.00

Flag question

A non-invasive perfusion MRI methodology used to quantify cerebral blood flow is

Select one:

- Functional MRI
- Single photon emission tomography
- Resting state connectivity
- Diffusion tensor imaging
- Arterial spin labelling

Explanation:

Arterial spin labelling is a magnetic resonance imaging perfusion technique that enables quantification of cerebral blood flow without the use of gadolinium contrast.

Ref: Ho ML. Arterial spin labeling: clinical applications. Journal of Neuroradiology. 2018 Sep 1;45(5):276-89.

The correct answer is: Arterial spin labelling

Question **57**

Not answered

Marked out of 1.00

Flag question

Which of the following questions is useful in evaluating immediate recall?

Select one:

- Can you tell me today's date?
- Where were you yesterday?
- What did you have for dinner?
- Where did you go to school?
- Can you repeat these six numbers?

Explanation:

Immediate memory refers to the amount of information that can be held in mind (like a phone number or digit span) to be available for immediate use.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 13.

The correct answer is: Can you repeat these six numbers?

Question **58**

Not answered

Marked out of 1.00

Flag question

Letter cancellation, star cancellation and line bisection tasks examine which of the following cognitive dysfunction?

Select one:

- Visual neglect
- Apraxia
- Agraphia

- Disorientation
- Acalculia

Explanation:

Visual neglect can be assessed with cancellation and line bisection tasks, in which the paper is placed at the patient's midline, and the patient is asked to either cross out stars/letters/lines on the page or bisect the single line presented.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 49.

The correct answer is: Visual neglect

Question **59**

Not answered

Marked out of 1.00

Flag question

REM pattern is recorded during the daytime in a student who has been studying for exams the last two weeks. The most likely reason is

Select one:

- Delirium
- Antipsychotic use
- Cocaine intoxication
- Sleep deprivation
- Normal pattern

REM sleep during daytime can be seen in narcolepsy, sleep deprivation or withdrawal from stimulants.

The correct answer is: Sleep deprivation

Question **60**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a component of clock drawing test?

Select one:

- Using numbers 1 to 12 to create the face of a clock
- Being aware of the current time when the test is administered
- Indicating the hour correctly according to instructions
- Indicating the minutes correctly according to instructions
- Placing the hands of the clock correctly

Explanation:

The clock-drawing test does not assess one's orientation to time.

Ref: https://www.sralab.org/sites/default/files/2017-07/Clock%20Drawing%20Test%20Instructions.pdf

The correct answer is: Being aware of the current time when the test is administered

Question **61**

Not answered

Marked out of 1.00

Flag question

Which of the following has a white, bright appearance in a CT scan?

Select one:

- White matter
- Haemorrhage
- Brain tumour
- Brain oedema
- CSF

Explanation:

The CT image is determined by the degree to which tissues absorb X-irradiation. The bony structures absorb high amounts of irradiation and tend to obscure details of neighbouring structures, an especially troubling problem in the brainstem, which is surrounded by a thick skull base. In a CT scan, CSF appears black, brain tissue appears grey, and the skull appears white. There is very poor discrimination between grey and white matter of the brain. Blood appears bright white. Certain tumours may be invisible on CT because they absorb as much irradiation as the surrounding normal brain. In comparison with MRI, the only component of the brain better seen on CT scanning is calcification, which may be invisible on MRI. T1 MRI images are the most similar to CT images. Blood appears bright in both CT and T1 MRI.

The correct answer is: Haemorrhage

Question **62**

Not answered

Marked out of 1.00

Flag question

Grey matter reduction is seen in patients with psychosis. The transition of patients from ultra high risk to psychotic patients involves a volume reduction in which part of the brain region?

Select one:

- Prefrontal cortex
- Caudate nucleus
- Cerebellum
- Superior temporal gyrus
- Hippocampus

Explanation:

Ultra high-risk patients that transition to psychosis have been reported to exhibit greater longitudinal volume reduction and/or cortical thinning in frontal areas (including orbitofrontal, superior frontal, middle frontal, and prefrontal cortices); temporal areas (such as the inferior and middle temporal cortex, and the fusiform and parahippocampal cortex); the medical and superior parietal lobes and precuneus; and the insular cortex.

Ref: Andreou C, Borgwardt S. Structural and functional imaging markers for susceptibility to psychosis. Molecular psychiatry. 2020 Nov;25(11):2773-85.

The correct answer is: Prefrontal cortex

Question **63**

Not answered

Marked out of 1.00

Flag question

Generalised diffuse delta and theta waves in the EEG are seen in

Select one:

- Temporal lobe seizures
- Metabolic encephalopathy
- Infantile spasms
- CJD
- Normal Stage 1 sleep

Explanation:

Diffuse slowing of the background rhythms of the EEG is the most common abnormality. It is non-specific and is present in patients with diffuse encephalopathies of diverse causes. The slowing of background activity in a metabolic encephalopathy usually begins with an increase in theta activity, followed by intermittent generalised theta slowing or bilateral synchronous delta waves groups, sometimes with a frontal maximum (frontal intermittent rhythmic delta activity).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 999.

Boutros NN, Galderisi S, Pogarell O, Riggio S. Standard electroencephalography in clinical psychiatry: A practical handbook. 2011. p. 72.

The correct answer is: Metabolic encephalopathy

Question **64**

Not answered

Marked out of 1.00

Flag question

Which of the following is an advantage of CT scans over MRI?

Select one:

- Gray vs. white matter demarcation is well delineated
- Posterior fossa lesions are well noted
- Multiplanar images are obtained
- Lesions close to bone are better seen
- Calcified lesions are better visualised

Explanation:

The CT image is determined by the degree to which tissues absorb X-irradiation. The bony structures absorb high amounts of irradiation and tend to obscure details of neighbouring structures, an especially troubling problem in the brainstem, which is surrounded by a thick skull base. In a CT scan, CSF appears black, brain tissue appears grey, and the skull appears white. There is very poor discrimination between grey and white matter of the brain. Blood appears bright white. Certain tumours may be invisible on CT because they absorb as much irradiation as the surrounding normal brain. In comparison with MRI, the only component of the brain better seen on CT scanning is calcification, which may be invisible on MRI.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 69-70.

The correct answer is: Calcified lesions are better visualised

Question **65**

Not answered

Marked out of 1.00

Flag question

A diabetic patient who is taking antidepressants is noted to have polyuria and polydipsia. His urine and plasma osmolality are both low. Choose the most likely cause for his presentation.

Select one:

- Psychogenic polydipsia
- Hyponatraemia
- Diabetes insipidus
- Hyperglycaemia
- SIADH

Explanation:

Psychogenic polydipsia is characterised by excessive fluid intake without an organic cause. A low plasma sodium concentration with a low urine osmolality is usually indicative of water overload due to psychogenic polydipsia. A high-normal plasma concentration in conjunction with a urine osmolality that is lower than plasma osmolality points towards diabetes insipidus.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 116.

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 269.

The correct answer is: Psychogenic polydipsia

Question **66**

Not answered

Marked out of 1.00

Flag question

During the course of a clinical interview, the psychiatrist says, 'So from what you have been telling me, your sleep is poor, you have excessive disturbing thoughts and you have lost weight in last 2 months. Is that right?' What technique is he using?

Select one:

- Interpretation
- Summation
- Facilitation
- Clarification
- Repetition

Explanation:

Summation can be used periodically during the interview in order to summarise what has been identified about a certain topic. This provides the opportunity for the patient to clarify or modify the psychiatrist’s understanding and possibly add new material.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 14.

The correct answer is: Summation

Question **67**

A 29-year-old African-American man who uses intravenous heroin presents with a puffy face and

Not answered

Marked out of 1.00

Flag question

steady increase in weight despite normal appetite. On examination his blood pressure is 150/100mmHg. The most likely diagnosis is

Select one:

- Hepatitis B liver disease
- Hypothyroidism
- Opiate withdrawal
- Nephropathy
- HIV seroconversion

Explanation:

"Heroin-associated nephropathy" is the term used to describe focal segmental glomerulosclerosis in the setting of heroin use. Patients present with proteinuria, which may be in the nephrotic range. It is more common among Black heroin users.

Ref: Sethi, S. The changing spectrum of heroin-associated kidney disease. CJASN. 2018: 975-976.

The correct answer is: Nephropathy

Question **68**

Not answered

Marked out of 1.00

Flag question

Inappropriate lack of concern about a functional deficit or disability is known as

Select one:

- Anosognosia
- Prosopagnosia
- La belle indifference
- Perseveration
- Alexithymia

Explanation:

(La) belle indifference is a surprising lack of concern for, or denial of, apparently severe functional disability. It is part of classical descriptions of hysteria and continues to be associated with operational descriptions of conversion disorder. It is also seen in medical illnesses (e.g. following a stroke) and is a rare and non-specific symptom of no diagnostic value.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 103.

The correct answer is: La belle indifference

Question **69**

Not answered

Marked out of 1.00

Flag question

Which of the following is not a functional neuroimaging technique?

Select one:

- fMRI scan
- Single positron emission computerised tomography

- Positron emission tomography
- Magnetoencephalography
- Digital subtraction angiography

Functional neuroimaging measures an aspect of brain function to understand the relationship between activity in certain brain areas and specific mental functions. Common methods of functional neuroimaging include SPECT, PET, fMRI, EEG and NIRS (near infra-red spectroscopic imaging). Digital Subtraction Angiography (DSA) is a powerful technique for the visualization of blood vessels in the human body. DSA is a fluoroscopy technique to visualize blood vessels in a bony or dense, soft tissue area. It does not measure brain function as such but identifies abnormalities in the integrity of cerebral vasculature.

The correct answer is: Digital subtraction angiography

Question **70**

Not answered

Marked out of 1.00

Flag question

A patient is asked to hold out both arms with palms facing upwards and maintain the posture with eyes closed. His left arm slowly drops down compared to right arm. Which of the following lesions is most likely to be present?

Select one:

- Left hemisphere lesion
- Cranial nerve X nuclei lesion
- Cerebellar vermis lesion
- Right cerebral hemisphere lesion
- Thoracic spinal cord lesion at T12

A pronator drift is elicited by having the patient extend both arms with the wrists supinated and palms facing upwards and eyes closed - slow unequal drift towards pronation indicates hemiparetic weakness. The left arm drop in this case indicates a right cerebral lesion.

The correct answer is: Right cerebral hemisphere lesion

Question **71**

Not answered

Marked out of 1.00

Flag question

Which of the following is not a frontal release sign?

Select one:

- Rooting reflex
- Glabellar tap
- Snout reflex
- Cremasteric reflex
- Grasp reflex

Explanation:

Primitive reflexes include the glabellar tap, rooting, snout, sucking, and palmomental reflexes. These

are generally absent in adults. When present in the adult, these signs signify diffuse cerebral damage, particularly of the frontal lobes (hence called frontal-lobe release signs). Cremasteric reflex is a superficial reflex.

Ref: Neurological History and Physical Examination - Medscape, http://emedicine.medscape.com/article/1147993-overview (accessed May 6, 2021)

The correct answer is: Cremasteric reflex

Question **72**

Not answered

Marked out of 1.00

Flag question

Clinical features of drug-induced hyponatraemia include

Select one:

- Muscle cramps
- Constipation
- Diarrhoea
- Tremors
- Sweating

Explanation:

Symptoms and signs of drug-induced hyponatraemia can range from nausea, muscle cramps/weakness, and malaise to hypertension, lethargy, confusion, and, if severe, seizures and coma.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 1026.

The correct answer is: Muscle cramps

Question **73**

Not answered

Marked out of 1.00

Flag question

Blood oxygen level dependent measurement BOLD is a technique used in which of the following imaging modalities?

Select one:

- MRS spectroscopy
- SPECT scan
- Functional MRI
- Contrast CT
- PET scan

Explanation:

The goal of fMRI is to visualise brain activity and so spatial resolution is sacrificed to the benefit of temporal resolution. fMRI is not at sufficient temporal resolution to directly measure the neuronal signal and so it measures an indirect index of neuronal activity – the blood oxygen-level dependent (BOLD) contrast. The principle behind BOLD is that an active population of neurons will receive more fresh, oxygen-risk blood flow than non-active neurons. It is this variation between the magnetic signal of oxygenated and deoxygenated blood in the brain that constitutes the basis for the fMRI signal.

Ref: Martinelli C, Shergill SS. Everything you wanted to know about neuroimaging and psychiatry, but were afraid to ask. BJPsych Advances. 2015 Jul;21(4):251-60.

The correct answer is: Functional MRI

Question **74**

Not answered

Marked out of 1.00

Flag question

A patient can follow commands and repeat phrases, but he has incomprehensible speech. He is likely to have

Select one:

- Transcortical aphasia
- Conduction aphasia
- Broca aphasia
- Wernicke aphasia
- Dyscalculia

Explanation:

When repetition is preserved in a patient with Broca (non-fluent) aphasia, it signifies transcortical motor aphasia, and the lesion is anterior to the Broca area. When repetition is preserved in Wernicke (fluent) aphasia, it is called transcortical sensory aphasia, and the lesion is posterior to the Wernicke area.

Ref: Neurological History and Physical Examination - Medscape, http://emedicine.medscape.com/article/1147993-overview (accessed May 6, 2021)

The correct answer is: Transcortical aphasia

Question **75**

Not answered

Marked out of 1.00

Flag question

The earliest appearance of brain atrophy in Alzheimer disease is most marked in which of the following brain regions?

Select one:

- Insular lobe
- Medial temporal lobe
- Frontal lobe
- Occipital lobe
- Posterior parietal lobe

Explanation:

Alzheimer disease is associated with **medial temporal lobe atrophy**, particularly of the **entorhinal cortex** and **hippocampus.**

Ref: McKhann, G.M. et al. 2011. The diagnosis of dementia due to Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's & dementia*, *7*(3), pp.263-269.

The correct answer is: Medial temporal lobe

Question **76**

Not answered

Marked out of 1.00

Flag question

A patient presents with sensory impairment over the contralateral face, arm, and leg involving pinprick, touch, vibration, position, two-point discrimination, and stereognosis. This patient is most likely to have a lesion in which of the following structures?

Select one:

- Frontal lobe
- Thalamocortical projections
- Hippocampal region
- Amygdalar nucleus
- Occipital lesion

Thalamo-parietal projections, when damaged, can give rise to the above reported sensory disturbances in addition to disturbances in higher sensory integration resulting in failed 2 point discrimination and stereognosis.

The correct answer is: Thalamocortical projections

Question **77**

Not answered

Marked out of 1.00

Flag question

Mr. X has been consuming a high dose of cannabis for a long time. How long will it be present in his urine?

Select one:

- 24 hours
- 10 days
- 3 days
- 7 days
- 4 weeks

Explanation:

Cannabis can be detected in urine after single use for up to 3 days. In the setting of chronic, heavy use, cannabis can be detected in the urine for up to 4 weeks.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 631.

The correct answer is: 4 weeks

Question **78**

Not answered

Marked out of 1.00

Flag question

Inability to recognise familiar faces is seen in

Select one:

- Anosognosia
- Simultanagnosia
- Prosopognosia

- Somatognosia
- Visuospatial apraxia

Explanation:

Prosopagnosia is the inability to recognise familiar faces that is not due to impaired visual acuity or level of consciousness.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1143.

The correct answer is: Prosopognosia

Question **79**

Not answered

Marked out of 1.00

Flag question

Which of the following is a supportive technique not aimed at eliciting information?

Select one:

- Clarification
- Redirection
- Validation
- Interpretation
- Symptom expectation

Validation helps to decrease a patient's sense of embarrassment about a feeling or behaviour. Validation is often employed when quoting how it is normal for people to have different emotions/ reactions/ behaviours, etc.

The correct answer is: Validation

Question **80**

Not answered

Marked out of 1.00

Flag question

Which of the following will you use to assess orientation in a hospitalised patient?

Select one:

- Ability to repeat a phone number immediately after reading out
- Tap the table once whenever the examiner reads letter 'E' from a random list of alphabets
- Ability to recognise pictures of famous personalities
- Ability to correctly identify the doctor when asked to point out
- Ability to remember his/her own address

Orientation to person is a stable domain which is generally lost only late in the course of a confusional state.

The correct answer is: Ability to correctly identify the doctor when asked to point out

Question **81**

Which of the following investigations involve introducing a synthetic radionucleotide for obtaining brain images?

Not answered

Marked out of 1.00

Flag question

Select one:

- T2 weighted scan
- PET scan
- fMRI
- T1 weighted scan
- Diffusion imaging

Explanation:

Both PET and SPECT are based on the idea that it is possible to learn about a dynamic process by introducing a measurable tracer into a target system and then tracking it as it travels around the system. Both PET and SPECT involve the administration of a radiopharmaceutical tracer which, once absorbed, emits gamma rays that can be detected by the scanner and thus provide an image of the distribution of the tracer in the body over time.

Ref: Martinelli C, Shergill SS. Everything you wanted to know about neuroimaging and psychiatry, but were afraid to ask. BJPsych Advances. 2015 Jul;21(4):251-60.

The correct answer is: PET scan

Question **82**

Not answered

Marked out of 1.00

Flag question

All of the following drugs will increase urine porphyrins in a susceptible individual with disordered haem metabolism except

Select one:

- Oestrogens
- Diclofenac
- Barbiturates
- Chlorpromazine
- Benzodiazepines

Acute intermittent porphyria is episodic in nature and the episodes are often triggered by certain medications including estrogens, barbiturates and benzodiazepines. Diclofenac can precipitate an episode. Psychiatric manifestations include depression, anxiety, delirium and psychosis.

The correct answer is: Chlorpromazine

Question **83**

Not answered

Marked out of 1.00

Flag question

The investigation that has no helpful role in diagnosing narcolepsy is

Select one:

- CSF analysis
- Human leucocyte antigen typing
- Multiple Sleep Latency test
- Sleep polysomnogram

- Sleep deprivation test

Explanation:

The sleep deprivation test is not helpful to diagnose narcolepsy.

Investigation	Findings in narcolepsy
Polysomnography (sleep EEG and Multiple Sleep Latency Test)	Sleep-onset REM period is highly-specific (25-50% of cases) Increased N1 (light sleep) and repeated awakenings
CSF hypocretin-1 levels	Levels <110pg/mL are highly specific and sensitive for narcolepsy with cataplexy (In 10%, levels may be normal or even high)
HLA typing	Strong association with HLA-DR2 haplotypes coded on chromosome 6 HLA DQB1*0602 and DQA1*0102 are found in up to 85-95% of individuals (compared with 12-38% in the general population)
Imaging	MRI useful to exclude some rare causes of secondary narcolepsy (abnormalities of the brainstem and diencephalon)
	Adapted from Oxford Handbook of Psychiatry, 2019. p. 451.

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Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 451.

The correct answer is: Sleep deprivation test

Question **84**

Not answered

Marked out of 1.00

Flag question

Which of the following speech best describes Broca aphasia?

Select one:

- Non-fluent aphasia
- Fluent aphasia
- Telegraphic speech
- Scanning speech
- Staccato speech

Explanation:

Broca aphasia (also called non-fluent or expressive aphasia) had traditionally been characterised by non-fluent speech and impaired repetition but intact auditory comprehension.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 48.

The correct answer is: Non-fluent aphasia

Question **85**

All of the following are signs of an upper motor neuron (UMN) lesion except

Not answered

Marked out of 1.00

Flag question

Select one:

- Spasticity
- Rigidity
- Muscular atrophy
- Extensor plantar
- Weakness

Explanation:

An upper motor neuron (UMN) lesion is a lesion anywhere from the cortex to the corticospinal tract. There are both positive and negative features of the UMN syndrome. Positive features include hyperreflexia, spasticity, and a positive Babinski reflex (presenting as an upward response of the big toe when the plantar surface of the foot is stored, with other toes fanning out). Negative features include impaired motor control, easy fatiguability, weakness, and loss of dexterity. Lower motor neuron (LMN) syndromes may arise from pathology affecting the distal motor nerve up to the level of the anterior horn cell. LMN lesions are characterised by muscle atrophy, weakness (like UMN lesions), and hyporeflexia without sensory involvement. Atrophy is seen in LMN lesions, not in UMN lesions.

Ref: Garg N et al. Differentiating lower motor neuron syndromes. Journal of Neurology, Neurosurgery & Psychiatry. 2017 Jun 1;88(6):474-83.

The correct answer is: Muscular atrophy

Question **86**

Not answered

Marked out of 1.00

Flag question

Which region of the brain is often highlighted when performing tasks inside a scanner during a functional MRI scan?

Select one:

- Thalamus
- Basal ganglia
- Cerebellum
- Primary motor cortex
- Hypothalamus

The region of the brain that is often highlighted by functional MRI scan is the primary motor cortex.

The correct answer is: Primary motor cortex

Question **87**

Not answered

Marked out of 1.00

Flag question

A 32-year-old man has left his wife and children without informing them, has traveled a long distance and has taken up a new name to avoid being found. He is drinking heavily, appears depressed and suicidal. Physical examination is unremarkable except for generalized lymph node enlargement and some muscle wasting. Which of the following is a likely diagnosis?

Select one:

- Major depression
- Alcohol dependence

- Dissociative fugue
- HIV illness
- Lymphoma

Generalised lymph node enlargement with muscle wasting along with social withdrawal (possibly due to perceived stigma) is suggestive of a HIV illness.

The correct answer is: HIV illness

Question **88**

Not answered

Marked out of 1.00

Flag question

T1 images in MRI are mostly useful for

Select one:

- Assessment of normal brain structure
- Visualising periventricular structures
- Visualising oedema
- Detecting tumours
- Localising ischaemic zones

Explanation:

The T1 image most closely resembles that of CT scans and is most useful for assessing overall brain structure. T1 is also the only sequence that allows contrast enhancement with the contrast agent gadolinium-diethylenetriamine pentaacetic acid.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 70.

The correct answer is: Assessment of normal brain structure

Question **89**

Not answered

Marked out of 1.00

Flag question

On mini-mental examination, a 74-year-old man loses all points on orientation to time and recall, while other faculties are intact. How many points has he lost in total?

Select one:

- Six
- Sixteen
- Thirteen
- Five
- Eight

5 points are awarded for orientation in time and 3 points for recall.

The correct answer is: Eight

Question **90**

Not answered

Marked out of 1.00

Flag question

Anomic dysphasia is suggestive of a lesion in which of the following regions of the brain?

Select one:

- Dominant fronto-temporal lesion
- Non-dominant fronto-temporal lesion
- Dominant temporo-parietal lesion
- Non-dominant temporo-parietal lesion
- Dominant occipital lesion

In anomic dysphasia, confrontation naming is affected more than any other function (when an object is shown, the patient cannot tell its name). It is often due to small left-sided lesions in language cortex - temporo-parietal junction.

The correct answer is: Dominant temporo-parietal lesion

Question **91**

Not answered

Marked out of 1.00

Flag question

Confrontation test carried out during physical examination is useful to detect which one of the following?

Select one:

- Astigmatism
- Myopia
- Cerebellar lesions
- Lens dislocation
- Lesions of optic tract

Explanation:

Confrontation test is used to assess visual fields with a wiggling finger or a red pin.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 128.

The correct answer is: Lesions of optic tract

Question **92**

Not answered

Marked out of 1.00

Flag question

A tremor that decreases with volitional movements and appears primarily when not engaged in tasks is most likely to be a

Select one:

- postural tremor
- cerebellar ataxia
- physiologic tremor
- senile tremor
- resting tremor

Explanation:

A resting tremor occurs in a body part that is relaxed and completely supported against gravity (e.g., when resting on the arm of a chair). It is typically enhanced by mental stress (e.g., counting backward) or movement of another body part (e.g., walking) and diminished by voluntary movement. Resting tremor is most commonly caused by parkinsonism, but may also occur in severe essential tremor.

Ref: Crawford III PF, Zimmerman EE. Tremor: sorting through the differential diagnosis. American family physician. 2018 Feb 1;97(3):180-6.

The correct answer is: resting tremor

Question **93**

Not answered

Marked out of 1.00

Flag question

An important investigation useful to differentiate Lewy body dementia from Alzheimer disease is

Select one:

- MRI scan
- CT scan
- X-ray
- EEG
- DAT scan

Explanation:

Reduced dopamine transporter (DAT) uptake in basal ganglia demonstrated by SPECT or PET imaging is an indicative biomarker of dementia with Lewy bodies. The utility of DAT imaging in distinguishing dementia with Lewy bodies from Alzheimer dementia is well-established, with sensitivity (78%) and specificity (90%).

Ref: McKeith IG et al. Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. Neurology. 2017 Jul 4;89(1):88-100.

The correct answer is: DAT scan

Question **94**

Not answered

Marked out of 1.00

Flag question

A patient who sustained a head injury reports loss of memory for events that happened between 13th September 2014 and 20th September 2014, though he retained full consciousness soon after the injury was sustained on 13th September 2014. Which of the following cognitive dysfunctions is he describing?

Select one:

- Semantic dementia
- Retrograde amnesia
- Selective amnesia
- Transient global amnesia
- Anterograde amnesia

Explanation:

Anterograde amnesia is the period of amnesia between an event (e.g. head injury) and the resumption of continuous memory. The length of anterograde amnesia is correlated with the extent of the brain injury.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 102.

The correct answer is: Anterograde amnesia

Question **95**

Not answered

Marked out of 1.00

Flag question

Which one of the following techniques can measure the extent of myelination of white matter fibres at a macroscopic level?

Select one:

- Resting state connectivity
- Diffusion tensor imaging
- Functional MRI
- Single photon emission tomography
- Arterial spin labelling

Explanation:

Diffusion tensor imaging (DTI) allows imaging of the organisation and orientation of brain white matter fibres by using the information derived from the movement of water molecules inside the tissues. DTI is an invaluable tool as it is at present the only method by which we can visualise white matter anatomy, which is still largely unknown.

Ref: Martinelli C, Shergill SS. Everything you wanted to know about neuroimaging and psychiatry, but were afraid to ask. BJPsych Advances. 2015 Jul;21(4):251-60.

The correct answer is: Diffusion tensor imaging

Question **96**

Not answered

Marked out of 1.00

Flag question

A patient agrees that he has significant problems with the use of alcohol. He takes responsibility and understands the importance of having a change in his lifestyle. He has not taken any active steps to change his drinking pattern so far and continues heavy alcohol use. Which of the following levels of insight does he have?

Select one:

- Intellectual insight
- Complete loss of insight
- Externalization of cause
- Emotional insight
- Awareness of problems but denial of responsibility

Intellectual insight: admission that the patient is ill and that symptoms or failures in social adjustment are caused by the patient's own particular irrational feelings or disturbances without applying this knowledge to future experiences.

The correct answer is: Intellectual insight

Question **97**

Not answered

Marked out of 1.00

Flag question

The brain region with most prominent changes observed using neuroimaging in ADHD is

Select one:

- Frontal cortex
- Occipital cortex
- Thalamus
- Parietal cortex
- Insula

Explanation:

Imaging studies have repeatedly shown underfunctioning of the frontostriatal pathways and underactivation of several prefrontal cortical areas, consistent with reduced cognitive executive functioning in ADHD.

Ref: Hill P. Attention-deficit hyperactivity disorder in children and adolescents: Assessment and treatment. BJPsych Advances. 2015 Jan;21(1):23-30.

The correct answer is: Frontal cortex

Question **98**

Not answered

Marked out of 1.00

Flag question

With a secluded patient, behavioural observation is carried out every 5th minute through a glass window. This technique is called

Select one:

- Event sampling
- ABC analysis
- Time sampling
- Functional analysis
- None of the listed options

Behavioural observation methods include observing and recording behavioural events, to study mental state or plan intervention. They are of 2 types: Event sampling: e.g. every fifth or tenth event is coded in detail. Time sampling: observations may be made only every 5 or 10 mins.

The correct answer is: Time sampling

Question **99**

Not answered

Marked out of 1.00

Flag question

If you perform a routine cognitive examination on a 90-year-old woman with no history suggestive of dementia, which of the following is least likely to be seen?

Select one:

- Reduced new learning efficiency
- Reduced working memory

- Reduced visuospatial ability
- Reduced motor speed
- Reduced attentional span

Reduced short-term, working memory is a feature of dementia. It is not expected as a part of healthy aging. But reduced motor speed, visuospatial skills, attentional span, and decreased new learning ability are seen in healthy aging process.

The correct answer is: Reduced working memory

Question **100**

Not answered

Marked out of 1.00

Flag question

Which of the following endocrine changes is noted in anorexia nervosa?

Select one:

- Increased luteinising hormone levels
- Decreased plasma cortisol levels
- Low thyroxine T4 levels
- Decreased carotene levels in plasma
- Raised growth hormone levels

In adolescents and adults with anorexia nervosa, increased basal growth hormone (GH) secretion, a higher number of secretory bursts, and higher GH concentration is seen. This can have adverse effects on bone mass and eventual growth curve.

The correct answer is: Raised growth hormone levels

Question **101**

Not answered

Marked out of 1.00

Flag question

Which of the following tests would be required to make a diagnosis of Huntington's disease in a patient without a family history?

Select one:

- EEG
- Immunological tests
- Sleep studies
- Gene testing
- Electromyography

Explanation:

The **diagnosis** of Huntington disease (HD) is straightforward if a patient with a **known family history** of HD presents with the typical neurological movement disorder (gene testing **not** required if there is laboratory confirmation of the Huntington disease gene in the family). In a patient **without a family history**, **gene testing is essential** but does not give a clinical diagnosis and must of combined with clinical history and physical exam.

Ref: Bachoud-Lévi AC et al. International Guidelines for the treatment of Huntington's Disease.

Frontiers in neurology. 2019;10:710.

The correct answer is: Gene testing

Question **102**

Not answered

Marked out of 1.00

Flag question

Which of the following clinical examination for apraxias is abnormal more commonly with a right than a left cerebral damage?

Select one:

- Imitating a toothbrush action
- Construction of polygons
- Mimicking hand movements
- Using tools appropriately
- Whistling and coughing on command

Constructional apraxia is inability to construct elements to a meaningful whole (e.g. inability to draw or copy simple diagrams or figures). This is typically caused by a lesion in the right cerebral hemisphere.

The correct answer is: Construction of polygons

Question **103**

Not answered

Marked out of 1.00

Flag question

In a motorbike accident, a 21-year-old driver sustains significant head injury. Following recovery from emergency neurological status, it is found that he has significant problems in error correction and categorization of objects. Which of the following is a likely area of lesion?

Select one:

- Diencephalon
- Parietal lobes
- Frontal lobes
- Occipital lobe
- Temporal lobe

Categorization and error correction are tests for frontal lobe functioning.

The correct answer is: Frontal lobes

Question **104**

Not answered

Marked out of 1.00

Flag question

A patient presenting with carbon monoxide poisoning is suspected to have bilateral posterior parietal damage due to anoxia. If this turns out to be a case of Balint syndrome, which of the following will NOT be a presenting feature?

Select one:

- Simultanagnosia
- Diplopia

- Oculomotor apraxia
- Optic ataxia
- Intact oculocephalic reflexes

Explanation:

Balint syndrome consists of a triad of simultanagnosia (inability to attend to more than one item of a complex scene at a time), optic ataxia (inability to guide reaching or pointing despite adequate vision), and oculomotor apraxia (inability to voluntarily direct saccades to a visual target). Fields may be full when challenged with gross stimuli, and oculocephalic reflexes are intact. This syndrome results from bilateral damage including the superior-parieto-occipital region, which disrupts the dorsal ('where') visual processing stream linking visual with parietal association areas. Possible causes include carbon monoxide poisoning, watershed infarction, leucodystrophy, and the posterior cortical variant of Alzheimer disease.

Ref: Cognitive assessment for clinicians -- Kipps and Hodges 76 .., http://jnnp.bmj.com/content/76/suppl_1/i22.full (accessed May 6, 2021)

The correct answer is: Diplopia

Question **105**

Not answered

Marked out of 1.00

Flag question

In which of the following conditions do you expect to see a pulvinar sign on magnetic resonance imaging?

Select one:

- Viral encephalitis
- Lewy body dementia
- Amyloid angiopathy
- Pick disease
- Prion disease

Explanation:

The pulvinar sign on MRI is diagnostic of variant CJD. It refers to **symmetric high-signal-intensity changes** affecting the **pulvinar and medial areas of the thalamus** and the **tectal plate** seen on **FLAIR sequence MRI.**

Ref: Ortega-Cubero S et al. Structural and functional neuroimaging in human prion diseases. Neurología. 2013 Jun 1;28(5):299-308.

The correct answer is: Prion disease

Question **106**

Not answered

Marked out of 1.00

Flag question

Which of the following is not a test for executive function?

Select one:

- Wisconsin Card Sorting Test
- Line orientation test
- Category Test

- Trail Making Test
- Stroop test

Explanation:

Visuospatial abilities (not executive function) can be assessed through procedures such as facial recognition and judgement of line orientation.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 54.

The correct answer is: Line orientation test

Question **107**

Not answered

Marked out of 1.00

Flag question

Which neuroimaging modality can give information about metabolic changes in the brain?

Select one:

- Structural MRI Scan
- PET scan
- Ultrasonogram brain
- SPECT scan
- CT scan

Explanation:

As with fMRI, SPECT provides information on cerebral blood flow, which is highly correlated with the rate of glucose metabolism, but does not measure neuronal metabolism directly. A key difference between PET and SPECT is that PET can measure glucose metabolism and blood flow as indices of neuronal activity, whereas SPECT can only measure blood flow. Structural abnormalities, not metabolic changes, are detected by structural MRI and CT. Ultrasound imaging involves sending high-frequency sound waves to a target region and registering their returning echoes – images can be captured in real time and so movement of blood flow, organs, or fetus can be captured as well.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 73.

Martinelli C, Shergill SS. Everything you wanted to know about neuroimaging and psychiatry, but were afraid to ask. BJPsych Advances. 2015 Jul;21(4):251-60.

The correct answer is: PET scan

Question **108**

Not answered

Marked out of 1.00

Flag question

In SPECT imaging iomazenil (I-123) is used to label

Select one:

- GABA-A receptors
- D-2 receptors
- Alpha-2 receptors
- 5-HT2 receptors
- Glycine receptors

Iomazenil is used for GABA-A receptors

The correct answer is: GABA-A receptors

Question **109**

Not answered

Marked out of 1.00

Flag question

A depressed patient who is having difficulties with alcohol use is attempting to narrate her problems regarding her job, but stops after initiating the discussion. The examiner praises the patient for doing well despite having a difficult time. Which of the following interview technique is he using?

Select one:

- Setting traps
- Reinforcement
- Acknowledgment of affect
- Suggestive lead
- Validation

Explanation:

Reinforcement interventions are very important in patients sharing material about themselves and other influential individuals and events. Without these reinforcements, often, the interview will become less productive.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 14.

The correct answer is: Reinforcement

Question **110**

Not answered

Marked out of 1.00

Flag question

A 18-year-old man presents with the bizarre behaviour of picking his clothes and the carpet. He appears agitated and mutters unintelligibly. He has a high temperature on examination. When examining him, it is found that when his knee is extended with flexed hip, he develops a 'catch' and flexes the other hip as a reflex. This is called

Select one:

- Reverse leg raising sign
- Positive Romberg's sign
- Straight leg raising sign
- Positive Brudzinki's sign
- Positive Kernig's sign

The Kernig's sign is elicited by flexing one hip and knee and then extending the knee with the hip still flexed. Hamstring spasm may occur; if severe, opposite knee may flex during the test - this is termed as a positive Kernig's.

The correct answer is: Positive Kernig's sign

Question **111**

The limbic system is implicated in a type of memory that deals with skills, habits, and nonassociative

Not answered

Marked out of 1.00

Flag question

learning and does not require conscious awareness and concentration, and usually remains intact after brain injury. This type of memory is called

Select one:

- Implicit memory
- Flash-bulb memory
- Insight memory
- Episodic memory
- Declarative memory

Explanation:

The performance of tasks such as typing, swimming, or cutting a loaf of bread are expressions of prior learning that differ from episodic memory in that there is no active awareness that memory is being searched in undertaking the particular skill. This type of memory is known as implicit, procedural, or skills memory. Studies of people with injury to the hippocampus suggest that declarative and implicit memory use different parts of the brain and can function independently. The hippocampus is believed to be the site where explicit (declarative) memory is stored, while implicit (procedural) memory is thought to reside in the limbic system, the amygdala, and the cerebellum.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 201 p. 64.

The correct answer is: Implicit memory

Finish review

PREVIOUS ACTIVITY
Classification

NEXT ACTIVITY
Descriptive Psychopathology

Jump to...

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